

Questions with Answer Keys

#MathBoleTohMathonGo

Q1 - Exponent of prime

The number of prime numbers among the numbers $105! + 2, 105! + 3, 105! + 4, \dots, 105! + 104, 105! + 105$, is

- (1) 31
- (2) 32
- (3) 33
- (4) None of these

Q2 - Exponent of prime

For some natural number N , the number of positive integral ' x ' satisfying the equation,

$$1! + 2! + 3! + \dots + (x!) = (N)^2, \text{ is}$$

- (1) one
- (2) two
- (3) infinite
- (4) None of these

Q3 - Restricted Arrangements

The number of arrangements of the letters of the word BANANA in which the two Ns do not appear adjacently, is

- (1) 40
- (2) 60
- (3) 80
- (4) 100

Q4 - Restricted Arrangements

How many different nine-digit numbers can be formed from the number 223355888 by rearranging its digits, so that odd digits occupy even positions, is

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

- (1) 16
- (2) 36
- (3) 60
- (4) 180

mathongo mathongo mathongo mathongo mathongo mathongo mathongo

Q5 - Restricted Arrangements

How many necklaces can be made of 18 beads, 8 being blue, 5 red, 4 green and 1 yellow, is

- (1) $\frac{(17)!}{2 \cdot 8!5!4!}$
- (2) $\frac{(17)!}{8!5!4!}$
- (3) $\frac{(17)!}{2 \cdot 5!4!}$
- (4) $\frac{(18)!}{2 \cdot 5!4!8!}$

mathongo mathongo mathongo mathongo mathongo mathongo mathongo

Q6 - Restricted Arrangements

The numbers of ways in which four S come consecutively in the word MISSISSIPPI, is

- (1) 8!
- (2) $\frac{8!}{4!}$
- (3) $\frac{8!}{4!2!}$
- (4) None of these

mathongo mathongo mathongo mathongo mathongo mathongo mathongo

Q7 - Restricted Arrangements

In how many ways three different rings can be worn in four figures with atmost one ring in each figure, is

- (1) 4
- (2) 24
- (3) 81
- (4) 64

mathongo mathongo mathongo mathongo mathongo mathongo mathongo

Q8 - Restricted Arrangements

mathongo mathongo mathongo mathongo mathongo mathongo mathongo

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

In how many ways five different rings can be worn in four figures with atleast one ring in each figure, is

- (1) 240
- (2) 6720
- (3) 480
- (4) None of these

Q9 - Geometric Applications

The number of lines drawn through 6 points lying on a circle, is

- (1) 12
- (2) 15
- (3) 24
- (4) 30

Q10 - Geometric Applications

The straight lines I_1, I_2, I_3 are parallel and lie in the same plane. A total number of m points are taken on I_1 , n points on I_2 , k points on I_3 . The maximum number of triangles formed with vertices at these points, are

- (1) ${}^{m+n+k}C_3$
- (2) ${}^{m+n+k}C_3 - {}^mC_3 - {}^nC_3 - {}^kC_3$
- (3) ${}^mC_3 + {}^nC_3 + {}^kC_3$
- (4) None of these

Q11 - Geometric Applications

Out of 18 points in a plane no three are in the same straight line except five points which are collinear. The number of straight lines that can be formed joining them, is

- (1) 143
- (2) 144
- (3) 153

Questions with Answer Keys

#MathBoleTohMathonGo

(4) None of these

Q12 - Geometric Applications

The maximum number of points of intersections of 8 straight lines, is

(1) 56

(2) 28

(3) 16

(4) 8

Q13 - Geometric Applications

There are 10 points in a plane of which no three are collinear and 4 points are concyclic the number of different circles that can be drawn through atleast 3 of these points, is

(1) 116

(2) 120

(3) 117

(4) None of these

Q14 - Geometric Applications

The maximum number of points into which 4 circles and 4 straight lines intersect, is

(1) 26

(2) 50

(3) 56

(4) 72

Q15 - Geometric Applications

The interior angles of a regular polygon measure 150° each the number of diagonals of the polygon, is

(1) 35

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

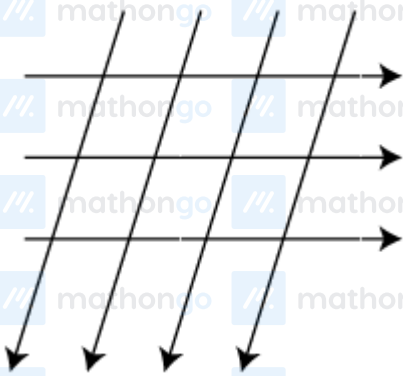
(2) 44

(3) 54

(4) 78

Q16 - Geometric Applications

The number of parallelograms in given figure, is



(1) 6

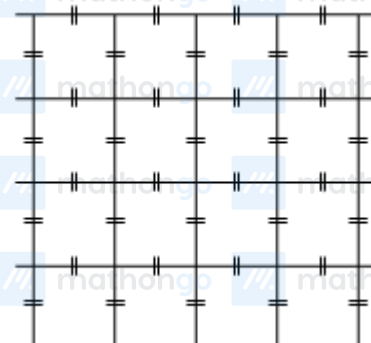
(2) 18

(3) 12

(4) 9

Q17 - Geometric Applications

The number of squares in given figure, is



(1) 8

(2) 18

#MathBoleTohMathonGo

Questions with Answer Keys

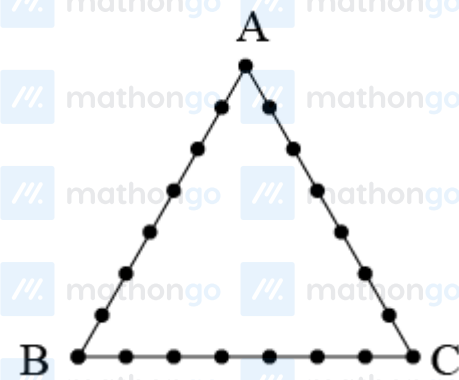
#MathBoleTohMathonGo

(3) 20

(4) 40

Q18 - Geometric Applications

18 points are indicated on the perimeter of a $\triangle ABC$ (see figure). How many triangles are there with vertices at these points



(1) 331

(2) 408

(3) 710

(4) 711

Q19 - Geometric Applications

Let there be 9 fixed points on the circumference of a circle. Each of these points is joined to every one of the remaining 8 points by a straight line and the points are so positioned on the circumference that atmost 2 straight lines meet in any interior point of the circle. The number of such interior intersection points, is

(1) 126

(2) 351

(3) 756

(4) None of the above

Q20 - Division in groups

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

The number of ways in which 9 students can be equally distributed among 3 sections, is

(1) $\frac{9!}{3(3!)^2}$

(2) $\frac{9!}{(3!)^3}$

(3) $9!$

(4) None of these

Q21 - Division in groups

The total number of ways of dividing 15 different things into groups of 8, 4 and 3 respectively, is

(1) $\frac{(15!)}{8!4!3!3!}$

(2) $\frac{(15!)}{8!4!3!}$

(3) $\frac{(15!)}{8!4!2!}$

(4) None of these

Q22 - Division in groups

In how many ways can we put 12 different balls in three different boxes such that first box contains exactly 5 balls, is

(1) $\frac{12!}{5!2^7}$

(2) $\frac{12!}{5!7!}$

(3) ${}^{12}C_5 \times 2^7$

(4) None of these

Q23 - Division in groups

The number of ways of distributing 50 identical things among 8 persons in which such a way that three of them get 8 things each, two of them get 7 things each and remaining 3 get 4 things each, is equal to

(1) $\frac{8!}{3!2!}$

(2) $\frac{8!}{(3!)^2 \cdot 2!}$

(3) $\frac{50!8!}{(3!)^2 \cdot 2!}$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(4) None of these

Q24 - Division in groups

The total number of different combinations of one or more letters which can be made from the letter of the word MISSISSIPPI, is

(1) 150

(2) 148

(3) 149

(4) None of these

Q25 - Division in groups

The number of ways in which 200 different things can be divided into groups of 100 pairs, is

(1) $\frac{(200)!}{2^{100}}$

(2) $\frac{(200)!}{(100)!2^{100}}$

(3) $(101) \cdot (102) \cdots (200)$

(4) None of these

Q26 - Division in groups

Seven different coins are to be divided amongst three persons. If no two of the persons receive the same number of coins but each receives atleast one coin and none is left over, then the number of ways in which the division may be made, is

(1) 420

(2) 630

(3) 710

(4) None of these

Q27 - Number of Divisors and Sum of divisors

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

How many divisors of 21600 are divisible by 10 but not by 15?

- (1) 10
- (2) 30
- (3) 40
- (4) None of these

Q28 - Number of Divisors and Sum of divisors

The sum of the divisors of $2^5 \cdot 3^7 \cdot 5^3 \cdot 7^2$, is

- (1) $2^6 \cdot 3^8 \cdot 5^4 \cdot 7^3$
- (2) $2^6 \cdot 3^8 \cdot 5^4 \cdot 7^3 - 2 \cdot 3 \cdot 5 \cdot 7$
- (3) $2^6 \cdot 3^8 \cdot 5^4 \cdot 7^3 - 1$
- (4) None of these

Q29 - Number of Divisors and Sum of divisors

Product of all even divisors of $N = 1000$, is

- (1) $32 \cdot 10^5$
- (2) $64 \cdot 2^{14}$
- (3) $64 \cdot 10^{18}$
- (4) $128 \cdot 10^6$

Q30 - Number of Divisors and Sum of divisors

The sum of all the numbers formed from the digits 1,3,5,7,9 which are smaller than 10000 if repetition of digits is not allowed, is

- (1) (28011)s
- (2) (28041)s
- (3) (28121)s
- (4) (29152)s, where $s = 25$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

Q31 - Dearrangement, Multinomial selection and arrangement

The number of ways can 10 letters be placed in 10 marked envelopes, so that no letters is in the right envelope are

(1) $10! \left\{ 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{1}{10!} \right\}$

(2) $10! \left\{ 1 + \frac{1}{1!} - \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{1}{10!} \right\}$

(3) $10! \left\{ 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{10!} \right\}$

(4) $9! \left\{ 1 + \frac{1}{1!} - \frac{1}{2!} + \frac{1}{3!} - \cdots - \frac{1}{10!} \right\}$

Q32 - Dearrangement, Multinomial selection and arrangement

The total number of integral solutions for the equation $x + y + z + t = 20$, where x, y, z and t are all ≥ -1 , is

(1) ${}^{27}C_3$

(2) ${}^{20}C_4$

(3) ${}^{23}C_3$

(4) ${}^{27}C_4$

Q33 - Dearrangement, Multinomial selection and arrangement

The number of integer solutions of $x + y + z = 0$ with $x \geq -5, y \geq -5, z \geq -5$, is

(1) 135

(2) 136

(3) 455

(4) 105

Q34 - Dearrangement, Multinomial selection and arrangement

The number of integral solutions of the equation $3x + y + z = 33$, where $x, y, z > 0$, is

(1) 210

Questions with Answer Keys

#MathBoleTohMathonGo

(2) 120

(3) 135

(4) 520

Q35 - Dearrangement, Multinomial selection and arrangementNumber of positive integral solutions satisfying the equation $x_1 \cdot x_2 \cdot x_3 \cdot x_4 \cdot x_5 = 240$, is

(1) 1875

(2) 1600

(3) 1250

(4) None of these

Q36 - Dearrangement, Multinomial selection and arrangement

The number of different words that can be formed out of the letters of the word MORADABAD taken four at a time, is

(1) 1500

(2) 600

(3) 620

(4) 626

Q37 - Dearrangement, Multinomial selection and arrangement

The number of words which can be formed by taking two same and two different letters from the letters of the word 'COMBINATION', is

(1) 756

(2) 1512

(3) 252

(4) None of these

Q38 - Dearrangement, Multinomial selection and arrangement

The total number of ways of selecting five letters from the letters of the word 'INDEPENDENT', is

- (1) 72
- (2) 3320
- (3) 120
- (4) None of these

Questions with Answer Keys

#MathBoleTohMathonGo

Answer Key

Q1 (4)	Q2 (2)	Q3 (1)	Q4 (3)
Q5 (1)	Q6 (3)	Q7 (2)	Q8 (3)
Q9 (2)	Q10 (2)	Q11 (2)	Q12 (2)
Q13 (3)	Q14 (2)	Q15 (3)	Q16 (2)
Q17 (3)	Q18 (4)	Q19 (1)	Q20 (2)
Q21 (2)	Q22 (3)	Q23 (2)	Q24 (3)
Q25 (2)	Q26 (2)	Q27 (1)	Q28 (4)
Q29 (3)	Q30 (2)	Q31 (1)	Q32 (1)
Q33 (2)	Q34 (1)	Q35 (4)	Q36 (4)
Q37 (1)	Q38 (1)		